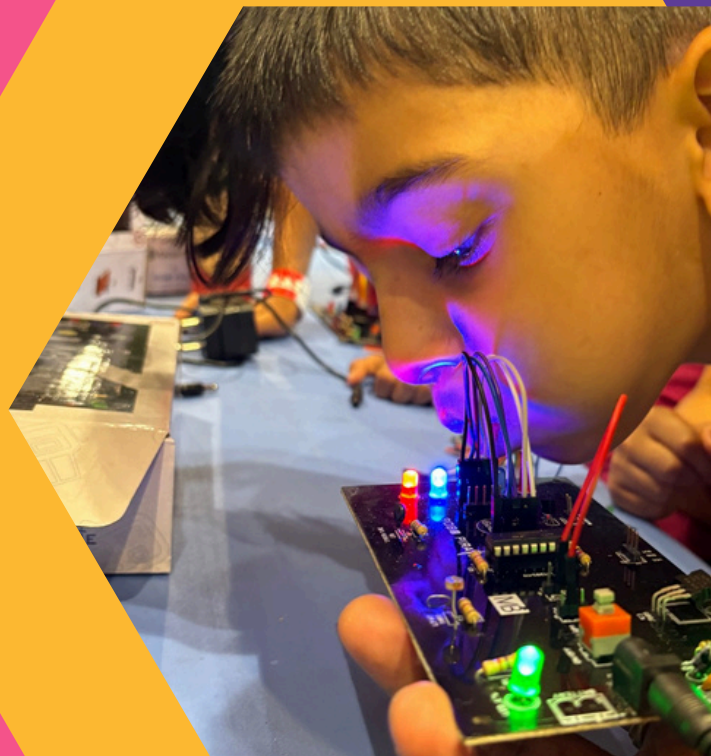


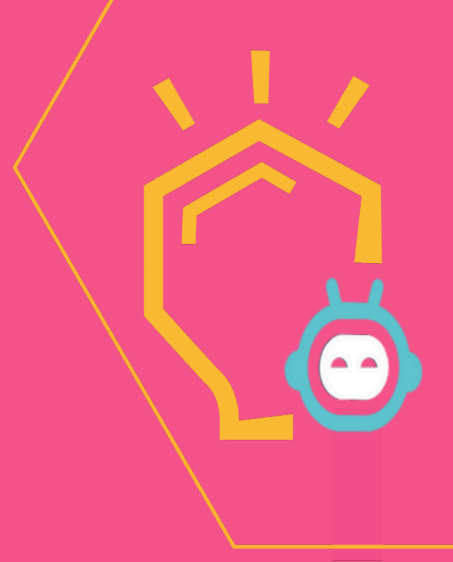


we **smart** **lab**

A complete, curriculum-aligned Robotics & AI lab



ABOUT US



WE Smart Lab: A Structured STEM Learning Ecosystem for Schools

WE Smart Lab is a **curriculum-driven Robotics & AI program** designed to integrate seamlessly into school academics.

The program emphasizes **hands-on, age-appropriate STEM** learning, enabling students to build strong conceptual foundations and real-world problem-solving skills.

Unlike short-term workshops, WE Smart Lab delivers **a year-long, grade-wise** curriculum supported by trained instructors, dedicated lab infrastructure, and continuous assessment.

Why Schools Choose WE Smart Lab

- **Curriculum-First Approach**

Grade-wise STEM curriculum for progressive skill development

- **Hands-On Learning**

Practical activities, DIY kits, and projects for every concept

- **Dedicated On-Site Instructor**

Ensuring consistent delivery and student safety

- **Measurable Outcomes**

Regular assessments and digital progress tracking

*Implemented across government and private schools in **6 states**.*

Everything Included in WE Smart Lab

A standardized, curriculum-first STEM lab model implemented across schools in multiple states.



Dedicated Smart Lab Setup



Dedicated On-site Instructor



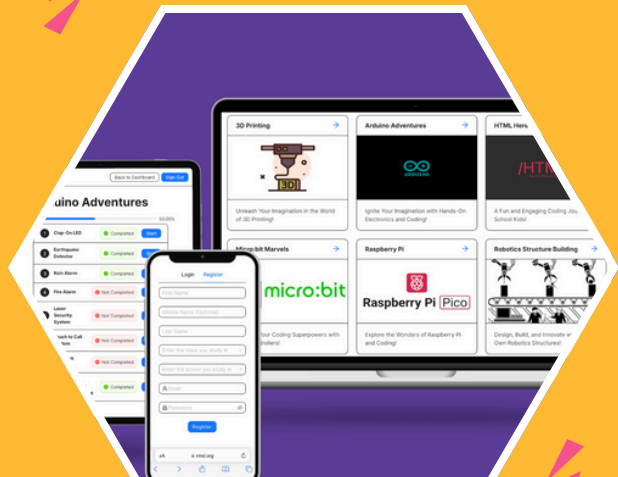
Grade-wise STEM Curriculum



Structured Weekly Sessions

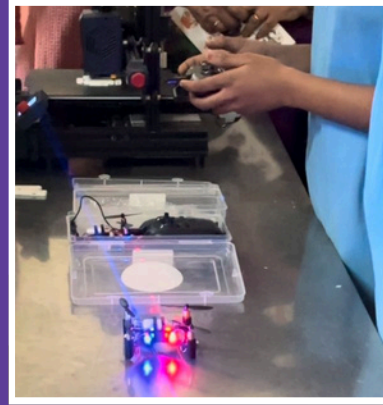
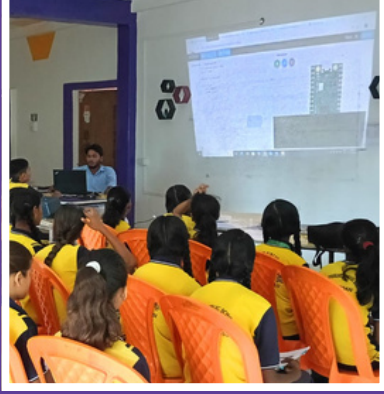


Hands-on Kits for Every Grade



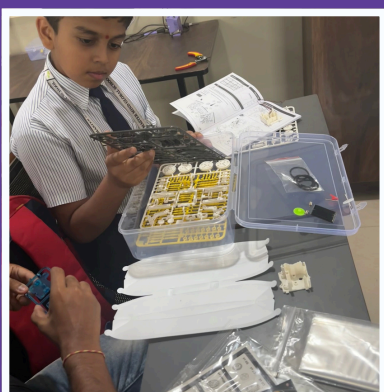
LMS & Digital Tracking System

*"Detailed implementation scope, roles, infrastructure, and operational responsibilities are clearly defined in the **WE Smart Lab BOM** for complete transparency and due diligence."*



Measurable Learning Impact Across Schools

85+ Labs established across India, impacting over 65,000 students through structured, curriculum-aligned STEM learning.



Masterboard

1

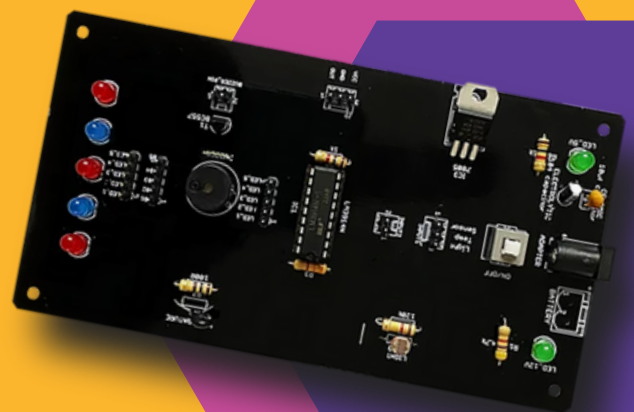
Key Learning Areas:

- Light & Dark sensing
- Temperature awareness
- Fire & smoke safety
- Touch & motion detection
- Sound and vibration
- Introductory logic through play

Sample Projects:

- Catch the Sneak
- Thirsty Plants

Sensing without programming! Have fun with Masterboard, where wires connect and lights blink like fireflies!



Microbit

2

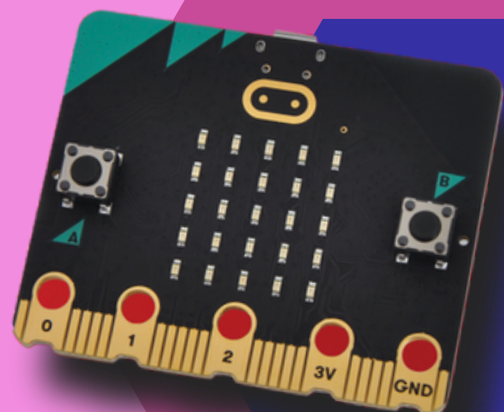
Key Learning Areas:

- Block-based visual programming
- Input/output using buttons and LEDs
- Logical sequencing and conditions
- Basic animations and games
- Sensor-based interactions
- Computational thinking through play

Sample Projects:

- Digital Dice
- Stopwatch
- Rock-Paper-Scissors
- Light Level Meter

Scratch-style coding made simple—students learn logic, sequencing, and problem-solving through visual programming.



Robotics Structure Building

3

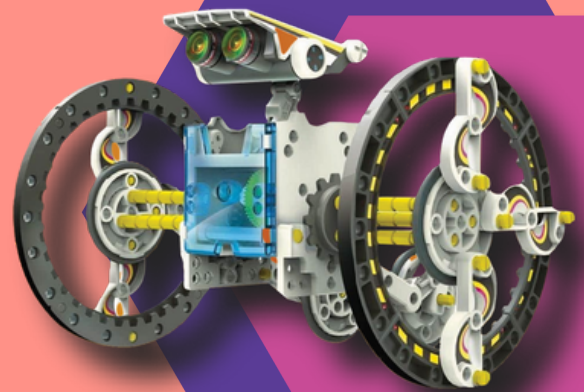
Key Learning Areas:

- Understanding robotic structures
- Mechanical assembly using gears, wheels, and frames
- Movement types: walking, rolling, rotating
- Balance, stability, and load handling
- Introduction to testing and iteration
- Problem-solving through build-test-improve cycles

Sample Projects:

- Walker Robot
- Beetle / Dog Robot
- Robo Vehicle

Building machines that move — before learning how to control them.



Basic Electronics & 3D Modeling

4

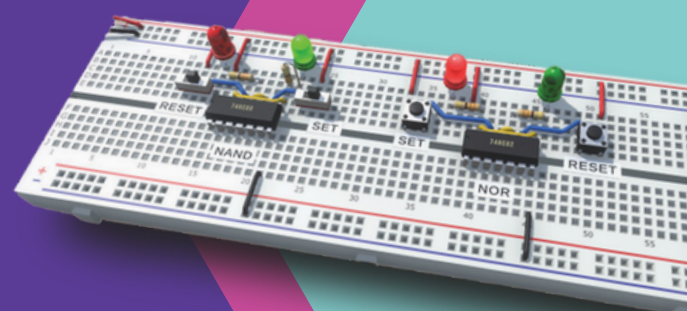
Key Learning Areas:

- Basics of electrical circuits and components
- LEDs, resistors, switches, and sensors
- Input/output behaviour and signal flow
- Breadboard prototyping and safe wiring
- Introduction to 3D modelling concepts
- Applying electronics to real-world use cases

Sample Projects:

- Light & Dark Sensor Circuit
- Electronic Buzzer / Alarm
- Simple Functional Prototype (with enclosure concept)

From moving structures to thinking systems — where logic meets design.



Mobile App Dev & Otto Bot

5

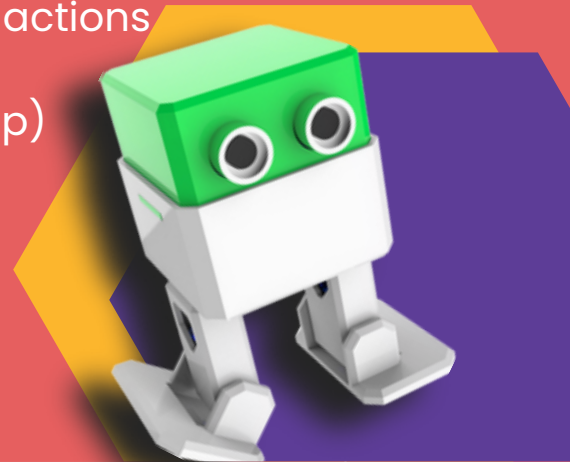
Key Learning Areas:

- Introduction to block-based app development
- User input handling and screen-based navigation
- Game logic and interactive app design
- Mechanical assembly of humanoid robots (Otto Bot)
- Servo motors, ultrasonic sensors, and movement control
- Connecting mobile apps with physical robot actions

Where students connect apps they build to robots they control.

Sample Projects:

- Simple Game App (Ball Bounce / Drawing App)
- Otto Bot Assembly & Movement
- App-Controlled Robot Interaction



Arduino Programming

6

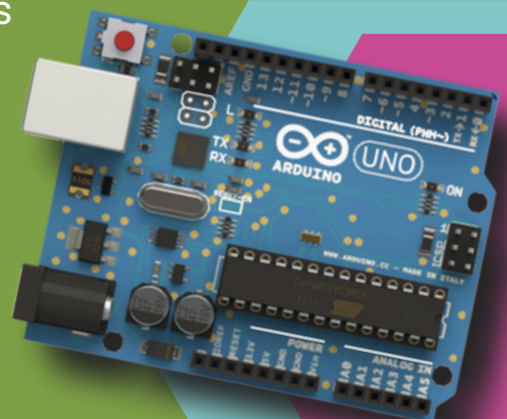
Key Learning Areas:

- Introduction to Arduino and embedded systems
- Input/output using LEDs, buzzers, sensors, and switches
- Reading sensor data and decision-based logic
- Servo motors and motion control
- LCD displays and user feedback systems
- Building real-world automation use cases

From coding logic to real-world automation using Arduino.

Sample Projects:

- LDR-Based Security Alarm
- Automatic Gate Opener
- Motion Detector Alarm System
- Digital Clock / Morse Code Decoder



LEO (AI Robotics & Intelligent Control)

7

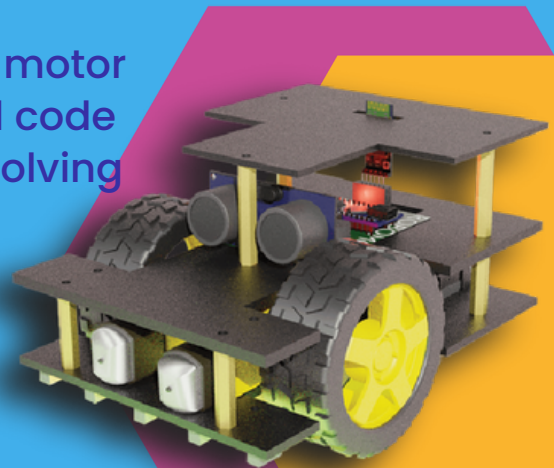
Key Learning Areas:

- Students design intelligent robots that make decisions
- Line following, obstacle detection, and path navigation
- Introduction to AI concepts through real-time decision making
- Working with IR sensors, ultrasonic, and motor
- Logic building using visual + text-based code
- Guided logic to independent problem-solving

Sample Projects:

- Line Following Robot
- Obstacle Avoidance & Maze Navigation
- Color Detection & Sorting Robot

Where students design intelligent robot behavior using sensors, logic, and real-time decision making.



HTML & Robotic Arm

8

Key Learning Areas:

- Fundamentals of web development using HTML & CSS
- Structuring web pages for real-world applications
- Connecting software interfaces with hardware systems
- Understanding servo motors and robotic arm kinematics
- Smartphone and browser-based robot control
- Applying logic to control real-world machines

Sample Projects:

- Web-Controlled Robotic Arm
- Smartphone-Controlled Pick & Place System
- Custom Servo Movement Sequences

Students connect web technologies with robotics to control machines through interfaces.



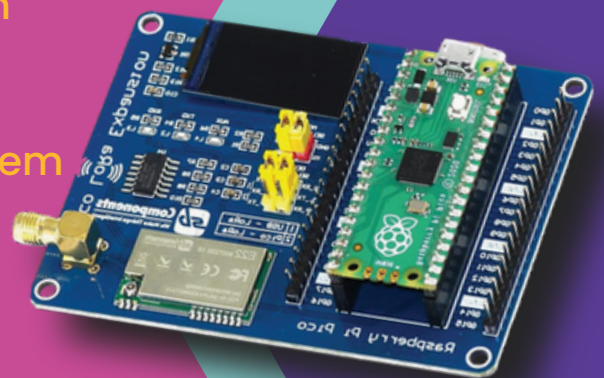
Key Learning Areas

- Introduction to Raspberry Pi and embedded computing
- Controlling LEDs, motors, and sensors through code
- Working with temperature, motion, and ultrasonic sensors
- Applying logic to real-world automation problems

Sample Projects

- Smart Fan & Sensor-Based Control System
- Automated Parking Gate

Students design real-world automation systems using Raspberry Pi and sensor-based logic.



Soldering Workshop



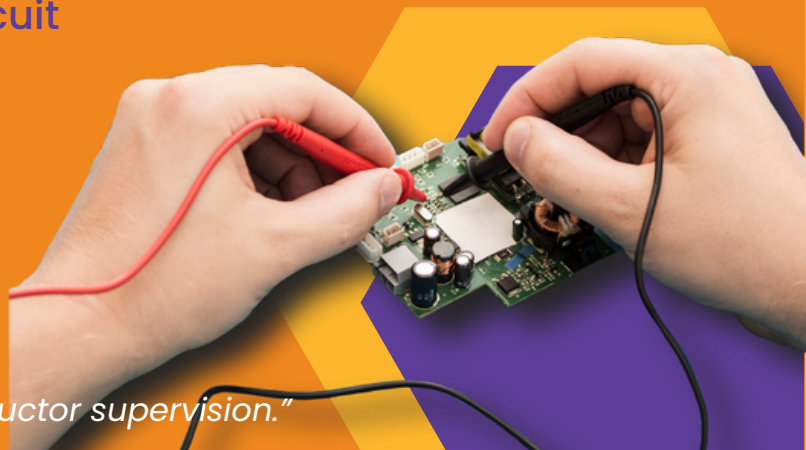
Key Learning Areas:

- Understanding electronic components and soldering tools
- Safe soldering practices and workstation setup
- Creating strong and clean solder joints
- Working with PCBs and basic circuit repair

Sample Activities:

- Resistor & Component Soldering Practice
- PCB-Based Circuit Assembly

Students learn safe and effective soldering techniques to build reliable electronic connections.





Drone Workshop



Key Learning Areas

- Basics of drone technology and components
- Mechanical assembly and motor–propeller setup
- Battery management and flight safety practices
- Understanding remote control and flight calibration

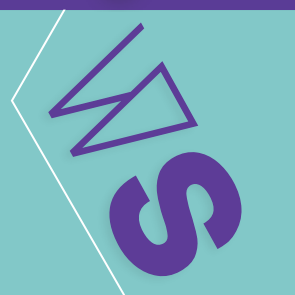
Students understand how aerial robots work by assembling, testing, and safely flying their own drones.

Sample Activities

- DIY Drone Assembly & Pre-Flight Testing
- Basic Flight Control and Maneuver Practice



Smart Home (IoT)



Key Learning Areas

- Understanding smart sensors and automation logic
- Input/output control using switches, relays, and sensors
- Environmental monitoring (temperature, rain, flame, motion)
- Applying IoT concepts to everyday use cases

Students build real-world automation systems by applying IoT concepts to everyday environments.

Sample Projects

- Smart Door, Alarm, and Temperature Control Systems
- Multi-Function Smart Home Automation Model



(Full chapter list available in teacher handbook)

**Educational
impact
achieved**

4+
Years of
structured
STEM program
delivery

85+
Labs
implemented
across India

65,000+
engaged through
hands-on STEM
learning

2 000+
completed
across grade
levels

NEP 2020 aligned | Skill-based education | Experiential learning

Why Schools Choose **WE Smart Lab**

Curriculum

- Grade-wise, curriculum-aligned STEM progression
- Integrated DIY kits, workbooks & LMS

Evaluation

- Weekly learning checks & periodic evaluations
- Academic year-aligned progress tracking

Portal

- Structured kits per grade (shared 1:2)
- Designed for repeatable classroom use

Support

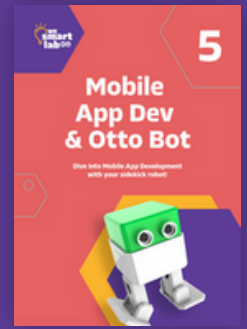
- Dedicated, full-time on-site trainer
- Continuous teacher coordination & safety oversight

DIY kits

- Centralized LMS for tracking & reporting
- Student progress, assessments & documentation

**Serving 6 states— shaping
futures with innovation
and quality!**

Implemented across government and private schools with consistent delivery standards.



WE Smart Lab

Empowering Tomorrow's Innovators

WE Smart Lab empowers students from Grades 1-9 through a structured, curriculum-driven STEM learning ecosystem.

We bring together grade-wise curriculum, hands-on kits, trained instructors, and digital tracking systems—all under one roof.

Through robotics, coding, electronics, and AI, we help students build strong foundational skills and thrive in an ever-evolving digital world.

Visit www.robopirate.in to discover how innovation and STEAM come together, adding a touch of magic to education in our community!

Schedule a school walkthrough or curriculum demo.

ROBO PIRATE

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